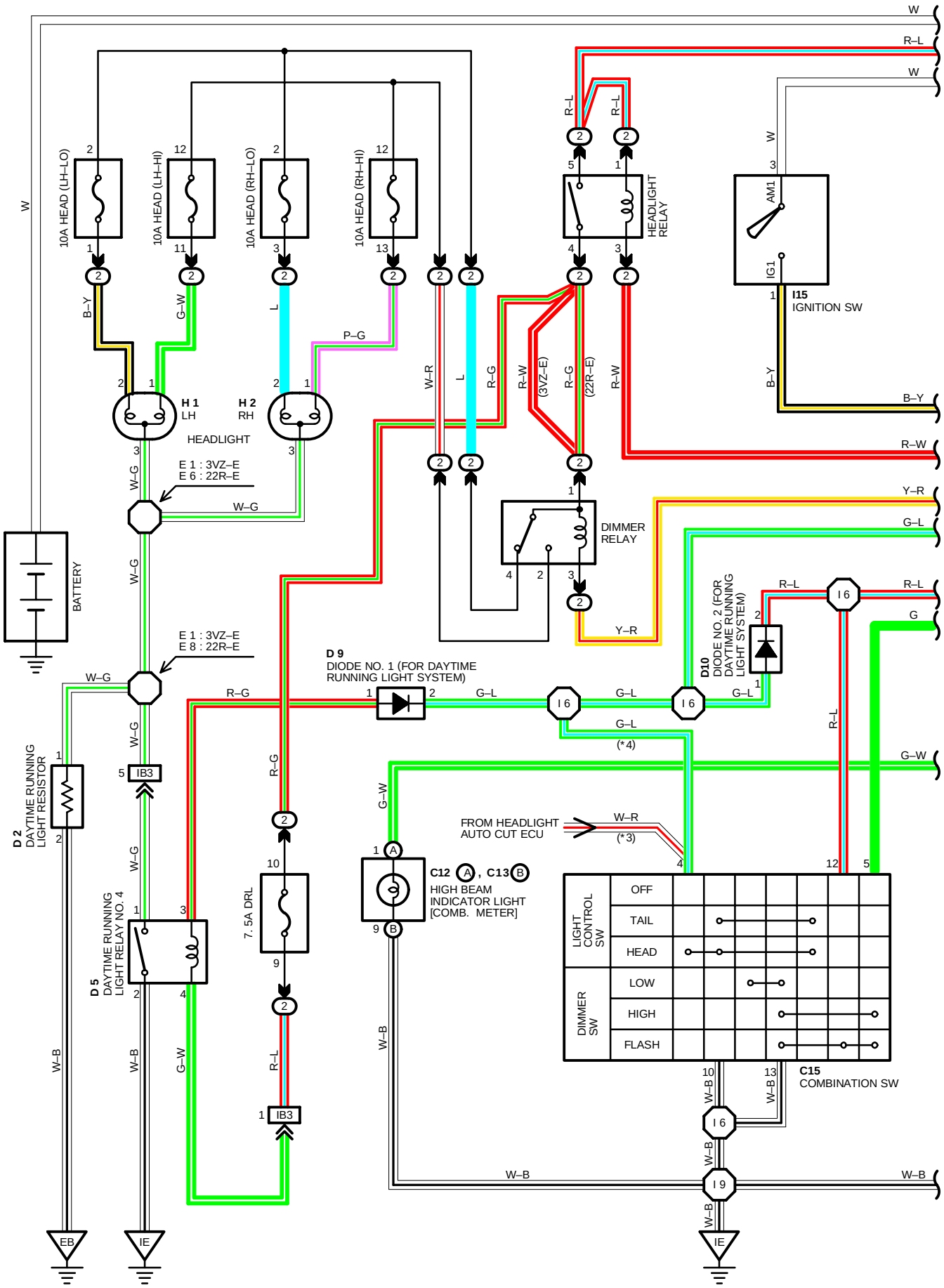
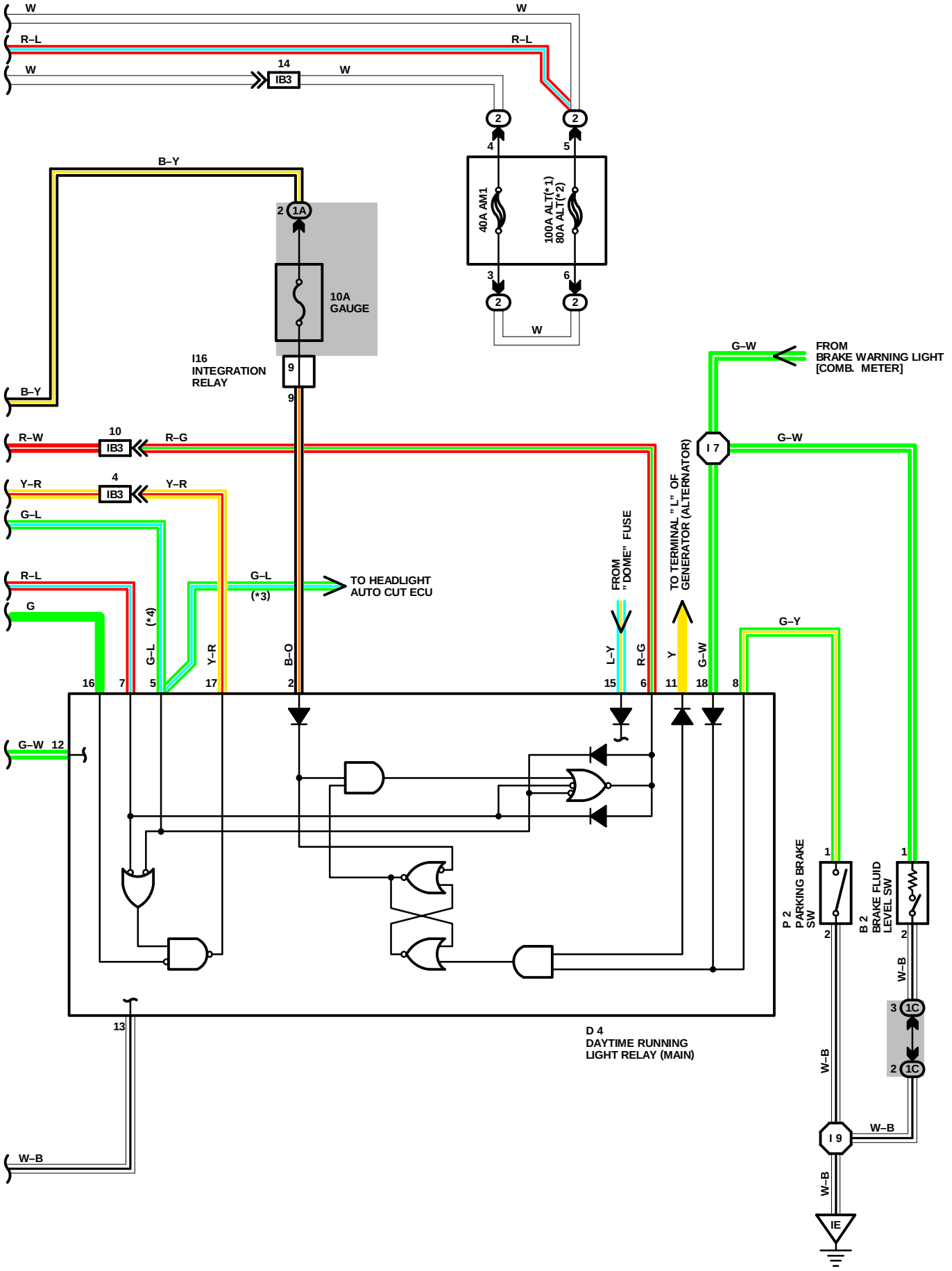
HEADLIGHT (FOR CANADA)



*1 : 22R–E A/T
*2 : 3VZ, 22R–E M/T
*3 : W/ HEADLIGHT AUTO CUT ECU
*4 : W/O HEADLIGHT AUTO CUT ECU



HEADLIGHT (FOR CANADA)

SYSTEM OUTLINE

1. DAYTIME RUNNING LIGHT OPERATION

WHEN THE ENGINE IS STARTED, VOLTAGE GENERATED AT **TERMINAL L** OF THE GENERATOR (ALTERNATOR) IS APPLIED TO **TERMINAL 11** OF THE DAYTIME RUNNING LIGHT RELAY (MAIN).

IF THE PARKING BRAKE LEVER IS PULLED UP (PARKING BRAKE SW ON) AT THIS TIME, THE RELAY IS NOT ACTIVATED SO THE DAYTIME RUNNING LIGHT SYSTEM DOES NOT OPERATE. IF THE PARKING BRAKE LEVER IS THEN RELEASED (PARKING BRAKE SW OFF), A SIGNAL IS INPUT TO **TERMINAL 8** OF THE RELAY (MAIN).

THIS ACTIVATES THE RELAY, SO CURRENT FLOWS FROM BATTERY TO THE HEADLIGHT RELAY (POINT SIDE) → **TERMINAL 1** OF THE DIMMER RELAY → **TERMINAL 4** → **HEAD (LH–LO)**, **(RH–LO)** FUSE → **TERMINAL 2** OF HEADLIGHTS → **TERMINAL 3** → **TERMINAL 1** OF DAYTIME RUNNING LIGHT RESISTOR → **TERMINAL 2** → **GROUND**, CAUSING THE HEADLIGHTS TO LIGHT UP AT 80%~85% OF THEIR NORMAL BRIGHTNESS.

ONCE THE DAYTIME RUNNING LIGHT SYSTEM OPERATES AND HEADLIGHT LIGHT UP, HEADLIGHTS REMAIN ON EVEN IF THE PARKING BRAKE LEVER IS PULLED UP (PARKING BRAKE SW ON).

IF THE ENGINE STALLS AND THE IGNITION SW REMAINS ON, HEADLIGHT REMAIN LIT UP EVEN THOUGH CURRENT IS NO LONGER OUTPUT FROM **TERMINAL L** OF THE GENERATOR (ALTERNATOR). IF THE IGNITION SW IS THEN TURNED OFF, THE HEAD GO OFF.

IF THE ENGINE IS STARTED WITH THE PARKING BRAKE LEVER RELEASED (PARKING BRAKE SW OFF), THE DAYTIME RUNNING LIGHT SYSTEM OPERATES AND HEADLIGHT LIGHT UP WHEN THE ENGINE STARTS.

2. HEADLIGHT OPERATION

DURING DAYTIME RUNNING LIGHT OPERATION, THE HEADLIGHTS ARE ON CONSTANTLY AT 80%~85% NORMAL INTENSITY.

WHEN THE LIGHT CONTROL SW [COMB. SW] IS SWITCHED TO **HEAD** POSITION, CURRENT FLOWS FROM THE **DRL** FUSE → **TERMINAL 4** OF DAYTIME RUNNING LIGHT RELAY NO. 4 → **TERMINAL 3** → **TERMINAL 1** OF DIODE NO. 1 → **TERMINAL 2** → **TERMINAL 4** OF LIGHT CONTROL SW [COMB. SW] → **TERMINAL 10** → **GROUND**, ACTIVATING DAYTIME RUNNING LIGHT RELAY NO. 4 CURRENT THEN FLOWS FROM THE **HEAD (LH–LO)**, **(RH–LO)** FUSE TO **TERMINAL 2** OF THE HEADLIGHTS → **TERMINAL 3** → **TERMINAL 1** OF DAYTIME RUNNING LIGHT RELAY NO. 4 → **TERMINAL 2** → **GROUND**, CAUSING THE HEADLIGHTS TO LIGHT UP AT NORMAL INTENSITY.

WHEN THE DIMMER SW [COMB. SW] IS SWITCHED TO THE **HIGH** POSITION, THE SIGNAL FROM THE DIMMER SW [COMB. SW] IS INPUT TO THE DAYTIME RUNNING LIGHT RELAY (MAIN). THIS ACTIVATES THE RELAY AND CURRENT FLOWS FROM **TERMINAL 1** OF THE DIMMER RELAY → **TERMINAL 3** → **TERMINAL 17** OF THE DAYTIME RUNNING LIGHT RELAY (MAIN), ACTIVATING THE DIMMER RELAY.

THIS CAUSES CURRENT TO FLOW FROM **TERMINAL 1** OF THE DIMMER RELAY → **TERMINAL 2** → **HEAD (LH–HI)**, **(RH–HI)** FUSE → **TERMINAL 1** OF HEADLIGHTS → **TERMINAL 3** → **TERMINAL 1** OF DAYTIME RUNNING LIGHT RELAY NO. 4 → **TERMINAL 2** → **GROUND**, CAUSING THE HEADLIGHTS TO LIGHT UP AT HIGH BEAM.

WHEN THE DIMMER SW [COMB. SW] IS SWITCHED TO **FLASH** POSITION, THE DAYTIME RUNNING LIGHT RELAY (MAIN) IS ACTIVATED AND CURRENT FLOWS FROM **TERMINAL 1** OF THE DIMMER RELAY TO **TERMINAL 2**. CURRENT FROM THE **DRL** FUSE FLOWS FROM **TERMINAL 4** OF DAYTIME RUNNING LIGHT RELAY NO. 4 → **TERMINAL 3** → **TERMINAL 1** OF DIODE NO. 1 → **TERMINAL 2** → **TERMINAL 1** OF DIODE NO. 2 → **TERMINAL 2** → **TERMINAL 12** OF DIMMER SW [COMB. SW] → **TERMINAL 13** → **GROUND**, AND ALSO FLOWS FROM THE **HEAD (LH–HI)**, **(RH–HI)** FUSE → **TERMINAL 1** OF HEADLIGHTS → **TERMINAL 3** → **TERMINAL 1** OF DAYTIME RUNNING LIGHT RELAY NO. 4 → **TERMINAL 2** → **GROUND**, CAUSING THE HIGH BEAM TO OPERATE.

WHEN THE HEADLIGHTS OPERATE (EXCEPT FOR FLASHING MODE), THE TAILLIGHTS ALSO OPERATE AS DESCRIBED IN PARTS 2.

SERVICE HINTS

HEADLIGHT RELAY

(2) 4– **(2)** 5 : CLOSED WITH LIGHT CONTROL SW AT **HEAD** POSITION OR DIMMER SW AT **FLASH** POSITION
: CLOSED WITH ENGINE RUNNING AND PARKING BRAKE LEVER RELEASED (PARKING BRAKE SW OFF)

DIMMER RELAY

CHANGED FROM HEAD(LO) TO HEAD(HI) WITH DIMMER SW AT **FLASH** POSITION OR WITH HEADLIGHT RELAY ON AND DIMMER SW AT **HIGH** POSITION

: PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
B 2	24 (3VZ–E)	D 2	26 (22R–E)	H 1	26 (22R–E)
	26 (22R–E)	D 4	28	H 2	24 (3VZ–E)
C12 A	28	D 5	28		26 (22R–E)
C13 B	28	D 9	28	I15	28
C15	28	D10	28	I16	28
D 2	24 (3VZ–E)	H 1	24 (3VZ–E)	P 2	28

: RELAY BLOCKS

CODE	SEE PAGE	RELAY BLOCKS (RELAY BLOCK LOCATION)
2	22	R/B NO. 2 (ENGINE COMPARTMENT RIGHT)

: JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1C	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)

: CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
IB3	34	ENGINE ROOM MAIN WIRE AND COWL WIRE (LEFT KICK PANEL)

: GROUND POINTS

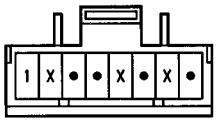
CODE	SEE PAGE	GROUND POINTS LOCATION
EB	30 (3VZ–E)	LEFT FENDER
	32 (22R–E)	
IE	34	LEFT KICK PANEL

: SPLICE POINTS

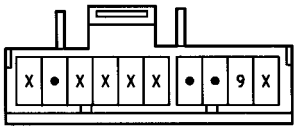
CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
E 1	30 (3VZ–E)	ENGINE ROOM MAIN WIRE	I 6	34	COWL WIRE
E 6	32 (22R–E)		I 7		
E 8	32 (22R–E)		I 9		

B 2 GRAY

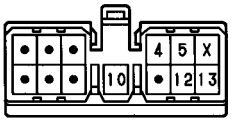
C12 (A)



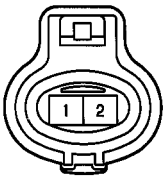
C13 (B) GRAY



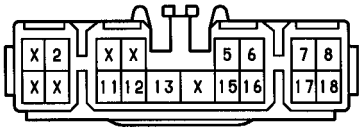
C15 BLACK



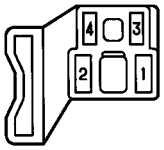
D 2 GRAY



D 4 GRAY



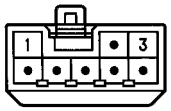
D 5



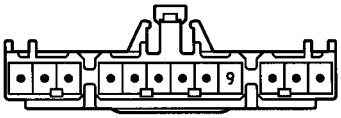
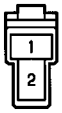
D 9, D10 BLACK

H 1, H 2 GRAY

I15 BLACK



I16



P 2